

is restricted due to potential interference with sensitive electronic fuzes or volatile materials.

By embedding Li-Fi modules into the facility infrastructure, DM Gosport maintains high-speed encrypted data exchange for mission command and inventory management while ensuring electromagnetic silence.

Europe and Africa – SWARM-C3 Project for maritime security. The SWARM-C3 (Command, Control, and Communication) project, led by Finland, represents a major step towards creating a combined command-and-control framework for autonomous unmanned underwater vehicles. Such vehicles serve as quiet sentries for naval security, as they can operate in disputed waters and remain difficult to detect. Swarm formations can organise complex manoeuvres and exchange mission data in real time using high-speed optical transceivers integrated into the vehicles.

While acoustic signals can be detected by adversaries, the blue-green Li-Fi light pulses are highly directional and dissipate rapidly. This allows the formation to remain hidden from long-range passive sonar while carrying out covert missions such as mine clearance or protecting subsea infrastructure.

NATO: Tactical optical network for secure battlefield communication. The tactical optical networks being developed by NATO use confined optical links to provide secure battlefield communication across mobile command posts and forward operating bases. The optical network complements existing communication layers and strengthens battlefield resilience by limiting vulnerability to jamming and interception.

NATO demonstrated Li-Fi's operational value during multinational exercises such as Noble Jump, during which optical systems were deployed in mobile command environments. Soldiers were able to



Li-Fi solutions being tested during NATO exercise (Source: <https://www.oledcomm.net>)



Li-Fi has the potential to deliver very safe, high-speed, and disguised communication links during combat (Source: <https://finabel.org>)

move between command enclaves without losing their secure, high-bandwidth connection—something conventional radio links cannot provide. During the exercise, Li-Fi also replaced kilometres of copper and fibre cabling previously used to connect command tents. This significantly reduced the physical load and logistical complexity associated with transporting and installing cables.

Li-Fi systems for defence: Key challenges

Li-Fi offers important advantages in terms of speed, safety, and flexibility when operating in environments exposed to electromagnetic threats. However, translating these advantages into effective defence operations requires overcoming several technical, environmental, and organisational challenges.

These challenges include maintaining stable performance and dependable operation under hostile

climatic conditions, mechanical vibration, and fluctuating ambient light. Reliable performance during critical missions is essential before the technology can be widely deployed in military environments.

A key limitation of Li-Fi is the requirement for a direct optical path between the transmitter and receiver. The performance of the link depends on whether the connection remains within line of sight, as unobstructed paths provide stronger signals.

Obstructions such as walls, armour plating, bulkheads, or even the movement of military personnel can interrupt the link. This adds complexity when

deploying Li-Fi systems in bunkers, armoured vehicles, and compartmentalised facilities where enclosed spaces require their own optical links.

Ongoing research is exploring multi-source lighting arrays that provide overlapping coverage, as well as reflective channels that redirect signals around obstacles. Beam-steering optics are also being developed to dynamically adjust transmission angles. Hybrid Li-Fi-RF systems are being tested to provide fallback connectivity when line of sight is disrupted.

Li-Fi has the potential to strengthen military operations by providing secure, high-speed, and covert communication links during combat. As the technology matures, it is likely to become an increasingly integral component of the communication infrastructure used by the Army, Navy, and Air Force, offering an operational advantage during critical situations. **EFY**